

Sagar Patel

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SUMMARY

Experienced Embedded Software Engineer with 2+ years at Cisco Systems and strong AI/ML research background. Currently advancing Generative AI and LLM expertise through NVIDIA certification. Proven track record in C/C++, Python development, and embedded systems optimization. Seeking opportunities in companies to leverage embedded AI convergence and autonomous systems expertise.

EXPERIENCE

Deep Learning Researcher | Omdena - Remote, Los Angeles, CA

Mar 2025 – Jun 2025

Project: Detecting Plastic Debris through Satellite Imagery in the Italian and Mediterranean Seas

- Developed and trained an optimized ResAttUNet deep learning model for binary segmentation of marine plastic debris proxy using Sentinel-2 imagery, achieving over 80% IoU accuracy.
- Curated a Mediterranean-specific dataset by merging Litter Windrows and MARIDA datasets, applying ACOLITE correction and class balancing via annular-region subsampling. Led spectral index filtering (NDVI, FDI) and patch-level sampling to enhance geospatial pattern recognition and reduce noise.
- Contributed modular, research-aligned code to the open-source community, advancing scalable plastic debris monitoring.

Software Developer | R-Tek – Remote, Los Angeles, CA

July 2024 – Present

- Integrated embedded hardware modules (Raspberry Pi, STM32, ESP32) with cloud services (IoT Core and MQTT) for real-time data acquisition, remote monitoring, and control.
- Developed firmware-to-cloud communication protocols over UART, I2C, and SPI, enabling IoT connectivity for edge devices.
- Automated testing frameworks for hardware-in-the-loop validation, improving system reliability and reducing manual debugging overhead.

Embedded Software Engineer | CISCO Systems Inc., Austin, TX

Jan 2023 – Apr 2024

- Led board bring-up and validation for next-gen enterprise switches (PHY, FPGA, PoE, TPM) achieving high hardware feature verification rate.
- Worked on C/C++ CLI tools with support for real-time voltage, thermal, and PoE telemetry.
- Automated board health monitoring via Python, cutting manual QA time and enabling daily Linux image testing pipelines.
- Collaborated with BIOS, Bootloader, Hardware, and QA teams to streamline board bring-up, tracking over 30+ hardware validation checkpoints.
- Used GitHub-based CI/CD to integrate user and kernel space drivers, ensuring 100% build reproducibility in test environments.

Firmware Engineer Intern | CISCO Systems Inc.

Jun 2022 – Sep 2022

- Prepared embedded firmware (C++) for PSE/PSU modules, optimizing runtime diagnostics and alerts with less response time.
- Expanded CLI features for switch power telemetry, increasing debugging efficiency in lab and field deployments.

PROJECTS

Development of Autonomous Drone for GPS-less Inspection | C++, Python

Dec 2022

- Developed a drone for autonomous inspection using Nvidia Jetson Nano, mRo Flight Controller, and Intel Realsense T265.
- Implemented SLAM (Simultaneous Localization and Mapping), Path Planning, and Obstacle Avoidance using ROS Noetic.
- Tested simulation in Gazebo and RViz, integrating real-time UI-based goal navigation.

SKILLS

Programming:	Python, C, C++, Bash & Shell Scripting, FreeRTOS
Technologies:	Embedded Systems, Linux Device Drivers, Firmware, Drones, Motion Planning, ROS 1
Hardware:	Raspberry Pi, Arduino, BeagleBone Black, STM32, ESP32, Nvidia Jetson Nano
Embedded Skills:	I2C, SPI, UART, Bluetooth, Multithreading, Inverters, BIOS, PWM Control
AI Skills:	PyTorch, Pandas, Computer Vision, Deep Learning (UNet, Attention Unet)
Design Software:	MATLAB, Simulink, Eclipse IDE, VS Code, Gazebo, QGroundControl
Embedded Software:	GDB, CPU Arch, Makefile, Cscope, Open Source, Middleware, CMake
Other Skills:	Docker, Git Version Control, GitHub, OpenCV, NumPy, CICD

EDUCATION

Sofia University

Master of Science in Computer Science

Costa Mesa, CA
Expected: Jun 2027

University of California, Irvine

Master of Embedded and Cyber Physical Systems (Computer Engineering) – GPA: 3.95/4.0

Irvine, California
Dec 2022

Charotar University of Science and Technology

Bachelor of Technology in Electronics and Communication Engineering – GPA: 4.0; Gold Medalist

Anand, India